

Data Science

Assignment – 10

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1 Using Machine Learning and Big Data to Analyse the Business Cycle (20 Points) On OLAT, you can find a link to an article by Hirschbühl et al. in which they outline how policymakers and decision makers employ data science techniques for the real-time assessment of economic activity. **1. Briefly summarise the four example scenarios ("Boxes") introduced in the paper. Describe the intended aim, dataset(s) used, and challenges encountered. Write between 300-400 words per scenario.**

BOX1:

The goal of Box 1 in the paper called "Monitoring labour market developments during the pandemic" is to give an overview of how the COVID-19 pandemic has affected the labour market in the Euro area. Data from Eurostat, the European Labour Force Survey (LFS), and the European Vacancy and Recruitment Report were used in this analysis (EVRR).

The pandemic has had a big effect on the job market, leading to a sharp rise in unemployment and a drop in the number of people who are working. Between February and November 2020, the number of unemployed people in the Euro area rose by 3.6 million, and the unemployment rate went from 7.5% to 8.3%. During the same time period, the number of people with jobs went down by 5.4 million, and the number of people in the labour force went down by 0.9%.

The article also says that the pandemic has hurt some groups of workers more than others. For example, young people and people with low skills are more likely to lose their jobs or have their hours cut. Also, the pandemic has had a bigger effect on women, who are more likely to stop working to take care of their families.

The paper talks about some of the problems that came up when trying to observe how the job market changed during the pandemic, especially when it came to the data and how it was analysed. Some of these difficulties are:

Difficulty in distinguishing between short-term and long-term effects of the pandemic on the labour market: The sudden and severe effects of the pandemic on the job market make it difficult to tell the difference between temporary and permanent changes in the job market.

Limited availability of data on certain aspects of the labour market: It is challenging to properly determine how the pandemic will affect these types of employees due to the lack of statistics on specific labour market factors, such as the number of people employed in the gig economy or under zero-hours contracts.

Data collection and processing: The pandemic has also caused problems with collecting and processing data, making it hard to get accurate and up-to-date information about how the job market is changing.

Quality of data: There may be differences in the quality of the data from different sources, which can be challenging to compare the data or draw conclusions.

Adjusting for the structural breaks: The sudden and severe changes in the labour market caused by the pandemic make it difficult to adjust for the structural breaks in the data, which can lead to unreliable estimates of labour market trends.

Even with these problems, the paper comes to the conclusion that it will be important to keep an eye on the labour market in order to understand how the pandemic is still affecting the labour market and to figure out which policy measures will help workers and businesses the most during these hard times.

BOX2:

Box 2 describes the development of a weekly economic activity tracker for the Eurozone. By combining monthly indicators with a long history of excellent forecast performance and high-frequency indicators that provide timely data, the tracker will provide a more reliable and accurate measure of regional economic activity. is intended for The authors caution that using high-frequency indicators can be problematic due to noise, complex seasonal patterns, and frequent data modifications. Additionally, the selection of indicators is not standardized, making the comparison of data between two countries very difficult.

The weekly economic activity tracker proposed in this article uses principal component analysis (PCA) with imbalanced data to address these issues. PCA is run twice. Once for the weekly series only and once for the weekly and monthly indicators. The first major component is used as a tracker that shows changes in activity from month to week on a weekly basis. The article states that the tracker performs relatively well and has a good balance of timely and reliable indicators, making it a useful tool for real-time tracking of economic activity.

The dataset used for the tracker contains several reliable monthly indicators with a long history of good forecast performance and timely high-frequency indicators. Indicators are selected based on several criteria. B. A sufficiently long history is available, not too noisy, and the metric weights in the all-combined aggregation. A major challenge in developing trackers was the short window of time available for high-frequency data. This makes it less reliable to build econometric relationships that prove to be stable over time compared to long time series of monthly economic indicators.

BOX3:

Machine learning techniques can be employed to make full use of big data findings. Box 3 talks about how economic sentiments such as PMI from newspaper articles can be used to predict GDP growth, especially in the 4 largest Euro Areas. These areas have a multilingual environment hence making it difficult to have one single source or dictionary.

There are two methods employed to build a sentiment analysis from text data.

1. Word count: Requires a pre-determined set of words for sentiment analysis. But the 4-euro areas have no common dictionaries. Translation to English may not be as effective if the dictionaries are not well constructed.
2. Semantic Clustering or Topic modelling: These methods are more complex and are based on extracting topics closer to sentiments.

Box 3 employs a word count approach which has a very high correlation with the Purchasing manager Index (PMI) from the year 2000-2019. Done by using an English dictionary and ignoring some noise, is successful in capturing the GDP growth during Recession. This, however, fails to encapsulate COVID-19 situation. It shows that the dictionary to be used relies on the nature of the the economic shock.

Studies have shown that prediction of Macroeconomic variables can be improved by text analysis. Linear methods are capable of making accurate predictions when there is no crisis situation. Machine learning can be employed to capture non-linearity in such situations and to filter out the noises.

Ridge regressions have lowest Root mean squared error and captures financial crisis better. Neural networks have given the most accurate results for the COVID-19 situation but were the opposite in predicting GDP growth during financial crisis.

This can be because of insufficient data available related to the financial crisis for training. Machine learning algorithms normally require a large amount of testing data for it to make highly accurate predictions.

BOX4:

Box 4 describes how big data and machine learning (ML) analysis can be used to measure economic policy uncertainty in the euro area and their impact on demand components using textual data from newspaper articles. The method presented in the text utilizes ML algorithms to classify text into different themes without prior knowledge, thereby avoiding selection bias. The article suggests that this approach can be applied to

various languages and can produce useful information for constructing stories and conducting economic analysis.

The study's main aim is to show how big data and machine learning can be used to measure economic policy uncertainty in the Eurozone and its impact on the economy.

To find sources of uncertainty using ML algorithms, the authors used datasets such as news articles from four Eurozone nations France, Germany, Italy and Spain over the period of 2000-2019. They discovered that the main causes of uncertainty include monetary, fiscal, political and trade policy, as well as European regulation and few others.

After that, they looked at the impact of these sources of uncertainty on private consumption and company investment using a structural vector autoregression approach. Increases in economic policy uncertainty, according to the study, had a detrimental effect on both private consumption and company investment. The article also points out that consumers and businesses may be affected differently by uncertainty resulting from numerous sources, such as energy, trade, and monetary policy. Due to space limitations, the study can only look at the effects of trade, monetary policy, and energy policy uncertainty.

Finding a suitable list of keywords, avoiding selection bias, and ensuring the ML algorithms can classify text into pertinent themes without prior knowledge are some of the difficulties that may have been encountered in this research.